The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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***Abstract*—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels within limited model and tool budgets. The Verified-Candidate Loop (VCL) lets validation evidence control candidate promotion. Its quality-of-results (QoR) score combines effective latency and worst-resource growth. QoR-guided retrieval-augmented generation (QoR-RAG) reuses compatible evidence, and Failure Reflection turns tool diagnostics into the next constraint. We evaluate three hosted large language model endpoints on a balanced 150-task suite. Targeted reruns replace records affected by application programming interface (API) failures. DeepSeek V4 Pro completes 147/150 tasks, Qwen3.5-122B-A10B completes 146. Qwen3.5 has the highest score sum across 115 scored tasks; DeepSeek has the highest conditional mean across 104. Qwen3.6 reaches 138/150 (92.0%) completion after targeted reruns, but only 67 tasks carry evaluator scores. Uneven scored subsets and residual provider failures prevent attributing endpoint differences to model weights alone.**

***Index Terms*—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results**

I. Evidence Must Control LLM Edits

Large language model (LLM) agents can edit unfamiliar high-level synthesis (HLS) kernels, but only tool reports certify correctness and hardware quality. A metered agent therefore needs a promotion rule that preserves a verified fallback when any required validation gate fails [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents also alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

Our Verified-Candidate Loop (VCL) contributes three artifacts and findings. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across seven gates: Interface, C simulation (CSim), HLS synthesis (Synth), Frequency, Capacity, required C/RTL co-simulation (CoSim), and Metric completeness. **(2) Hardware objective.** The bounded score Q_{HW} rejects speedups that lose more clock or worst-resource quality than they gain in latency. **(3) Endpoint evaluation.** Section IV separates task completion from score coverage across three hosted endpoints. DeepSeek leads completion coverage, Qwen3.5 has the largest scored set and highest score sum, and Qwen3.6 retains a structural repair gap. Uneven scored subsets and residual provider failures prevent attributing endpoint differences to model weights alone.

II. Tool Evidence Controls the Agent Workflow

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding hidden tests and frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs seven gates in order: Interface, CSim, Synth, Frequency, Capacity, required CoSim, and Metric completeness. Frequency enforces 100 MHz, Capacity enforces U55C limits, and CoSim applies only when the task requires it. For candidate c, define validity $V(c)$ over applicable gates $G_i \in \{0, 1\}$; an inapplicable CoSim gate contributes one:

$$V(c) = G_{\mathrm{int}} G_{\mathrm{csim}} G_{\mathrm{syn}} G_{\mathrm{freq}} G_{\mathrm{cap}} G_{\mathrm{cosim}} G_{\mathrm{metric}}. \tag{1}$$

Let b denote the verified fallback. The promotion rule is

$$P(c) = V(c) \wedge [Q_{\mathrm{HW}}(c) > Q_{\mathrm{HW}}(b)]. \tag{2}$$

Budget admission reserves the credits for all required gates. A failed gate or budget denial retains b. QoR-RAG supplies version-compatible rules, successes, and measured failures. Failure Reflection converts bounded diagnostics into the next constraint without another LLM request. Duplicate candidates and repeated failures stop retries. Hidden acceptance remains an evaluator-side check.

Figure 1 summarizes the loop. The three campaigns record 691 candidate interface checks, 112 interface rejections, 487 synthesis gates, and 67 CoSim gates. The audit establishes contract execution but does not isolate mechanism effects.

III. One Score Rejects False Speedups

The evaluator selects a valid starter as anchor a and falls back to a frozen reference hidden from the agent. Let c denote a candidate, L its cycle latency, $p = \max(p_{\mathrm{target}}, p_{\mathrm{estimated}})$ its effective period, and II its initiation interval. The score uses the II branch only when the task marks II relevant and both values are positive:

$$r_L = \frac{p_a L_a}{p_c L_c}, \quad r_P = \begin{cases} r_L^{0.85} (II_a / II_c)^{0.15}, & \text{reliable } II, \\ r_L, & \text{latency only}, \end{cases}. \tag{3}$$

For anchor and candidate resource counts A_j and C_j, let $\mathcal{R}$ contain lookup tables (LUTs), flip-flops (FFs), digital

Task and budget →	Candidate and evidence →	Validation chain Interface/CSIM/SYNTH/Frequency/Capacity	→ Promote or retain
Kernel and interface →	LLM + QoR-RAG →	required COSIM/Metric completeness	→ Verified fallback
→	← Failure Reflection from measured diagnostics		

Fig. 1. **Measured evidence controls every state transition.** A candidate replaces the verified fallback only after every applicable gate passes and Q_{HW} improves.

signal processing (DSP) blocks, block RAM (BRAM), and UltraRAM (URAM). The worst growth defines area quality:

$$\mathcal{R}_{+} = \{j \in \mathcal{R} : A_j > 1 \vee C_j > 1\},$$
$$\widehat{\mathcal{R}} = \begin{cases} \mathcal{R}_{+}, & \mathcal{R}_{+} \neq \varnothing, \\ \mathcal{R}, & \mathcal{R}_{+} = \varnothing, \end{cases}$$
$$r_A = \left[\max_{j \in \widehat{\mathcal{R}}} \frac{\max(C_j, 1)}{\max(A_j, 1)} \right]^{-1}. \tag{4}$$

The evaluator score combines validity, bounded hardware utility, and budget efficiency:

$$h = r_P^{0.55} r_A^{0.45}, \qquad Q_{\mathrm{HW}} = 1 - \frac{1}{(1+h)^2}, \tag{5}$$
$$E = \max\big(0.80,\, 1 - 0.10 u_{\mathrm{credit}} - 0.10 u_{\mathrm{time}}\big),$$
$$S = 100 V Q_{\mathrm{HW}} E. \tag{6}$$

Here the two u terms denote consumed fractions clipped to $[0, 1]$. Promotion compares Q_{HW} only after $V(c) = 1$, so remaining budget never turns an inferior design into the fallback.

IV. Completion Outpaces Score Coverage

Protocol. Three hosted endpoints process the same balanced 150-task suite under one agent revision and 9,000-credit limit. Each completed output must pass CSIM, SYNTH, 100 MHz, U55C capacity, and every applicable COSIM gate under the Track-A contract [8]. For tasks whose original records contain failed API requests, the merged corpus selects the available targeted-rerun record and recomputes completion counts. "Scored" counts tasks with a non-null evaluator score; the reruns do not add scores for every recovered completion. The merged records retain 1, 0, and 4 failed requests, respectively. Appendix A gives task provenance; Appendix C gives category and selected-record accounting details.

TABLE I
Reruns recover completion, while score coverage remains uneven. "Scored mean" is conditional on the scored set; "API fails" counts failed requests in the merged records.

Endpoint	Completed	Scored	Scored mean	Score sum	API fails
DeepSeek V4 Pro	147/150	104	71.95	7483.07	1
Qwen3.5-122B-A10B	146/150	115	68.26	7850.21	0
Qwen3.6-27B	138/150	67	66.90	4482.31	4

Results. DeepSeek provides the broadest and most balanced coverage: 147/150 completions and at least 24/25 in every category. Qwen3.5 trails by one completion, but its 115 scored tasks form the largest scored set and produce the highest score sum, 7850.21. DeepSeek's higher scored mean, 71.95, applies to a different 104-task subset and therefore does not establish a paired quality advantage.

Qwen3.6 reaches 138 completions, including 25/25 in compile repair, synthesis repair, and QoR optimization. Its residual gap concentrates in structural repair (18/25) and code generation (21/25). Only 67 Qwen3.6 tasks carry evaluator scores, so its score sum does not measure most recovered completions. Uneven scored subsets and residual provider failures prevent attributing endpoint differences to model weights alone.

V. Limits and Next Steps

The VCL makes validation evidence and Q_{HW} the promotion authority. The endpoint evaluation separates completion coverage from available score evidence. Uneven scored subsets and residual provider failures prevent attributing endpoint differences to model weights alone. Targeted reruns reduce the infrastructure confound but do not replace repeated campaigns. Matched ablations and broader corpora must test causal mechanism effects and transfer.

References

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Appendix A
Benchmark Sources and Task Composition

The accepted manifest freezes every upstream path, commit, task description, public test, evaluator-only hidden test, reference, and gate configuration. The builder rejected construction candidates that failed acceptance checks and retained 150 tasks with no cross-category kernel overlap. Table II separates six capabilities and exposes weak repair modes hidden by an aggregate success rate.

The kernels come from two public AMD repositories at fixed commits. The introductory examples contribute 109 tasks under Apache-2.0; the acceleration examples contribute 41 under MIT. Controlled faults define generation and repair conditions, while frozen valid starters or evaluator-only references anchor QoR evaluation. The 150 variants reuse 65 unique source paths. The source repositories are https://github.com/Xilinx/Vitis-HLS-Introductory-Examples and https://github.com/Xilinx/Vitis_Accel_Examples.

Appendix B
Track-A Requirement Mapping and Scope

Table IV distinguishes runtime-output evidence from submission-package obligations. "Strict" refers to fail-closed enforcement: missing evidence for an applicable gate rejects the output. Strict enforcement allows inapplicable gates to remain unexecuted. All tasks require CSim and Synth, whereas the manifest requires CoSim for the 25 structural tasks. The public guideline also says to pass CoSim without defining task-level exemptions. This report therefore scopes its CoSim evidence to the structural tasks and does not claim CoSim coverage for the remaining 125 tasks. Organizer confirmation or broader evidence must close this submission requirement.

The campaigns use the three checkpoints recommended by the public guideline: DeepSeek V4 Pro, Qwen3.5-122B-A10B, and Qwen3.6-27B. The guideline records their architecture and quantization; the hosted provider build remains opaque.

Appendix C
Detailed Three-Endpoint Results

All campaigns cover the same 150 tasks and share one source hash and credit limit. Section IV reports completion, score coverage, and conditional scores. Targeted reruns increase completion without adding evaluator scores to every recovered task. The comparison retains one selected record per task.

The category table reveals the residual pattern. DeepSeek completes at least 24 tasks in every category. Qwen3.5's four misses concentrate in generation and functional repair. Qwen3.6 completes every compile, synthesis, and QoR task, while structural repair remains lowest at 18/25. Its 67 scored tasks also cover less than half of the corpus, so score-based comparisons describe a narrower subset than completion-based comparisons.

Required ablations. A matched rerun should add one row per agent variant and report campaign identifier, success, repeated failures, mean Q_{HW}, tokens, calls, credits, and wall time. Planned variants are full agent, no QoR-RAG, no measured-failure retrieval, and no Failure Reflection. Until such runs exist, we print no value and claim no causal improvement.

Appendix D
Reproduction and Result Refresh

The accepted task list is `tasks/track_a_150/candidate_manifest.json`. The three source report paths appear as comments at the end of `technical-paper/results_generated.tex`. After replacing or extending matching directories under `runs/`, execute `python3 technical-paper/scripts/update_results.py` from the repository root. The script selects the latest matching final report and refuses to emit macros unless model identity, task count, API client, execution-source hash, and credit limit agree.

The generated file supplies task counts, result prose values, and all detailed table rows. A refreshed campaign updates numerical evidence without hand-editing the paper. Evaluator-only tests and references remain outside the agent-visible filesystem; report writers redact credentials and sensitive paths.

The campaigns use source snapshot `0a06af39777b6ae7f3962afa2910232eaf782e91727e0f184ec168f1a41f106c`, temperature 0, a 4,096-token output limit, a 180-second request timeout, and up to two retries after the first request. OpenRouter identifies the requested open-weight model and license mapping. The opaque provider build remains unfrozen. The run directory names retain campaign timestamps.

TABLE II
The test suite balances six agent capabilities. Each category contains 25 tasks; only the structural category requires C/RTL co-simulation in the accepted manifest.

Task type	Initial condition	Required output	Tasks	CoSim
Code generation	Signature-only stub	Complete kernel	25	0
Compile repair	Compile failure	Compiling kernel	25	0
Synthesis repair	Synthesis failure	Synthesizable kernel	25	0
Functional repair	CSim failure	Functionally correct kernel	25	0
Structural repair	CoSim failure	Executable RTL structure	25	25
QoR optimization	Valid baseline	Better valid Q_{HW}	25	0

TABLE III
Fixed source revisions make the corpus reconstructable.

Repository	License	Upstream commit	Tasks
Xilinx/Vitis-HLS-Introductory-Examples	Apache-2.0	`aa5c160faf5d5ebf58674df8f0591f9984ebae0f`	109
Xilinx/Vitis_Accel_Examples	MIT	`81187602355a7c2b666351154c5acca2074cae64`	41

TABLE IV
The run contract maps the applicable Track-A evaluation requirements to recorded evidence. Submission files and video remain author-side checklist items.

Track-A requirement	Enforcement and recorded evidence
U55C; Vitis 2025.2	The container fixes the target part and tool release; reports preserve the execution-source revision.
CSim, Synth, applicable CoSim	A missing or failed required gate rejects the candidate. Hidden tests and frozen references remain evaluator-only.
At least 100 MHz; feasible resources	Frequency and U55C capacity gates precede promotion. Reports retain latency, II, period, LUT, FF, DSP, BRAM, and URAM.
Budget and token accounting	Each run report records tokens and credits. Structured reports retain request, tool, and wall-time details. The campaign limit is 9,000 credits.
Recommended model comparison	OpenRouter campaigns cover DeepSeek V4 Pro, Qwen3.5-122B-A10B, and Qwen3.6-27B under one manifest and source revision. Mock, replay, and scripted backends cause the result generator to stop.
Docker and reproducibility	The repository supplies the Docker workflow, public task materials, structured reports, and result-generation script.
PDF, source archive, video	Authors separately confirm IEEE format, the two-page boundary, archive contents, and the five-minute demonstration as package obligations.

TABLE V
Category completion and QoR scores expose different endpoint profiles. Completion entries give tasks out of 25 and the corresponding rate. The QoR mean covers the same 25 tasks and uses Eq. (6).

Endpoint	Generate	Compile	Synth	Functional	Structural	QoR completed	QoR mean
DeepSeek V4 Pro	24/25 (96%)	25/25 (100%)	25/25 (100%)	24/25 (96%)	24/25 (96%)	25/25 (100%)	80.49
Qwen3.5-122B-A10B	22/25 (88%)	25/25 (100%)	25/25 (100%)	24/25 (96%)	25/25 (100%)	25/25 (100%)	76.55
Qwen3.6-27B	21/25 (84%)	25/25 (100%)	25/25 (100%)	24/25 (96%)	18/25 (72%)	25/25 (100%)	76.43

TABLE VI
Selected task records retain token and credit fields. Token counts are in millions. For a targeted-rerun task, rerun tokens replace the original token value, whereas credits remain from the original task record. The rows therefore neither estimate additive campaign spend nor support cost-efficiency comparisons.

Endpoint	Selected-record tokens (M)	Original-record credits
DeepSeek V4 Pro	3.36	2,398
Qwen3.5-122B-A10B	4.85	2,650
Qwen3.6-27B	4.18	1,805